

Decision and Social Choice

PE3101P — Lecture 4: Expected utility theory

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Today

- 1 Probability
- 2 Utility
- 3 Preference
- 4 Prospects
- 5 Axioms
- 6 The Expected Utility Theorem
- 7 Meaning
- 8 Objections

Housekeeping

- ▶ Reminder: Assignment 1 due **Tuesday 8th September**, by 23:59.
- ▶ Slides now go up **before** the lecture. Follow along on your own screen if it helps.
- ▶ Week 1 optional exercise solutions are on the website.
- ▶ TE1 tutorial: **Friday 10:00, AS2-0311.**

Conditional Credences Are Credences

Proposition

Suppose Cr satisfies the probability axioms and $\text{Cr}(B) > 0$. Define

$$\text{Cr}_B(A) = \text{Cr}(A \mid B).$$

Then Cr_B satisfies the probability axioms too.

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Then Cr_B satisfies the probability axioms too.

- ▶ So everything we know about credences carries over after conditioning.
- ▶ Three axioms, three things to check.

Non-Negativity, Conditioned

$$\text{Cr}(A \mid B) \geq 0 \text{ ?}$$

What we want.

Non-Negativity, Conditioned

$$\frac{\text{Cr}(A \cap B)}{\text{Cr}(B)} \geq 0 ?$$

Substitute the definition of $\text{Cr}(A \mid B)$.

Non-Negativity, Conditioned

$$\frac{\text{Cr}(A \cap B)}{\text{Cr}(B)} \geq 0$$

$\text{Cr}(A \cap B) \geq 0$ by non-negativity, and $\text{Cr}(B) > 0$. A non-negative number divided by a positive one.

Non-Negativity, Conditioned

$$\frac{\text{Cr}(A \cap B)}{\text{Cr}(B)} \geq 0$$

So Cr_B is non-negative.

Normalisation, Conditioned

$$\text{Cr}(\Omega \mid B) = 1 ?$$

What we want.

Normalisation, Conditioned

$$\frac{\mathbf{Cr}(\Omega \cap B)}{\mathbf{Cr}(B)} = 1 ?$$

Substitute the definition.

Normalisation, Conditioned

$$\frac{\mathbf{Cr}(B)}{\mathbf{Cr}(B)} = 1 ?$$

$\Omega \cap B = B$, since B is already inside Ω .

Normalisation, Conditioned

$$\frac{Cr(B)}{Cr(B)} = 1$$

Anything positive divided by itself.

Normalisation, Conditioned

$$\frac{Cr(B)}{Cr(B)} = 1$$

So Cr_B is normalised.

Finite Additivity, Conditioned

Take $X \cap Y = \emptyset$.

$$\text{Cr}(X \cup Y \mid B) = \text{Cr}(X \mid B) + \text{Cr}(Y \mid B) ?$$

What we want.

Finite Additivity, Conditioned

Take $X \cap Y = \emptyset$.

$$\text{Cr}(X \cup Y \mid B)$$

Start from the left-hand side.

Finite Additivity, Conditioned

Take $X \cap Y = \emptyset$.

$$\frac{\text{Cr}((X \cup Y) \cap B)}{\text{Cr}(B)}$$

Substitute the definition.

Finite Additivity, Conditioned

Take $X \cap Y = \emptyset$.

$$\frac{\text{Cr}((X \cap B) \cup (Y \cap B))}{\text{Cr}(B)}$$

Intersection distributes over union.

Finite Additivity, Conditioned

Take $X \cap Y = \emptyset$.

$$\frac{\mathbf{Cr}(X \cap B) + \mathbf{Cr}(Y \cap B)}{\mathbf{Cr}(B)}$$

$X \cap B$ and $Y \cap B$ are disjoint, so finite additivity applies to $\mathbf{Cr}$.

Finite Additivity, Conditioned

Take $X \cap Y = \emptyset$.

$$\frac{\text{Cr}(X \cap B)}{\text{Cr}(B)} + \frac{\text{Cr}(Y \cap B)}{\text{Cr}(B)}$$

Split the fraction.

Finite Additivity, Conditioned

Take $X \cap Y = \emptyset$.

$$\text{Cr}(X \mid B) + \text{Cr}(Y \mid B)$$

The definition, twice, in reverse.

Finite Additivity, Conditioned

Take $X \cap Y = \emptyset$.

$$\text{Cr}(X \mid B) + \text{Cr}(Y \mid B)$$

So Cr_B is finitely additive. All three axioms hold.

Expectation

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- ▶ A die: $E[X] = 1 \cdot 1/6 + 2 \cdot 1/6 + \dots + 6 \cdot 1/6 = 3.5$.
- ▶ Expected utility is the expectation of the random variable “utility of the outcome”.

Conditional Expectation

Definition

Where $x_1, \dots, x_n$ are the possible values of X and $\text{Cr}(B) > 0$,

$$E[X \mid B] = \sum_{j=1}^n x_j \cdot \text{Cr}(X = x_j \mid B).$$

- The same formula, run with the conditioned credences.

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- ▶ The same formula, run with the conditioned credences.
- ▶ Those are credences, as we showed. So this is a genuine expectation: the expectation of X under Cr_B .

The Law of Total Expectation

Law of total expectation

Where $B_1, \dots, B_m$ partition Ω , each with $\text{Cr}(B_i) > 0$:

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- ▶ The credence-weighted average of the conditional expectations.
- ▶ Compare total probability: $\text{Cr}(A) = \sum_i \text{Cr}(B_i) \text{Cr}(A \mid B_i)$.

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- Roll a die; X is the number shown. Split Ω into *odd* and *even*.

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- ▶ So $E[X] = \text{Cr}(\text{odd}) \cdot 3 + \text{Cr}(\text{even}) \cdot 4 = 1/2 \cdot 3 + 1/2 \cdot 4 = 3.5$.
- ▶ The formula looks complicated. It isn't.

Proving the Law of Total Expectation

$$\sum_i \text{Cr}(B_i) E[X \mid B_i]$$

Start from the right-hand side.

Proving the Law of Total Expectation

$$\sum_i \text{Cr}(B_i) \sum_j x_j \text{Cr}(\{X = x_j\} \mid B_i)$$

Substitute the definition of $E[X \mid B_i]$.

Proving the Law of Total Expectation

$$\sum_i \sum_j x_j \mathbf{Cr}(B_i) \mathbf{Cr}(\{X = x_j\} \mid B_i)$$

Move $\mathbf{Cr}(B_i)$ inside the inner sum.

Proving the Law of Total Expectation

$$\sum_i \sum_j x_j \mathbf{Cr}(\{X = x_j\} \cap B_i)$$

Definition of conditional probability, in reverse:

$$\mathbf{Cr}(B_i) \mathbf{Cr}(A \mid B_i) = \mathbf{Cr}(A \cap B_i)$$

Proving the Law of Total Expectation

$$\sum_j x_j \sum_i \text{Cr}(\{X = x_j\} \cap B_i)$$

Swap the order of the two sums.

Proving the Law of Total Expectation

$$\sum_j x_j \mathbf{Cr}(X = x_j)$$

The B_i partition Ω , so the inner sum is $\mathbf{Cr}(X = x_j)$ by total probability.

Proving the Law of Total Expectation

$$E[X]$$

The definition of $E[X]$.

Proving the Law of Total Expectation

$E[X]$

Which is the left-hand side.

The Thing We Never Defined

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What is utility?

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- ▶ **Utility as value.** Utility to the extent that the outcome is actually valuable — prudentially, morally, or in another sense. MEU becomes MEV: maximise expected value.

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What Utility Is, Continued

- ▶ **Utility as degree of desire.** How much the agent wants the outcome, all things considered.
- ▶ **Utility as a pure representative device.** No psychological connection with the agent: a mathematical representation of preferences she already has.
- ▶ Words can mean what we want them to. I use “utility” for the representative device. That is today’s sense.

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BDRC

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- ▶ **False.** Psychological hedonism is a thesis about what *motivates* people.
- ▶ The MEP Principle is a thesis about what people *should* do: what they do insofar as they are rational.
- ▶ So the two are independent. Either could be true while the other is false.

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BDRC, p. 76

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- ▶ If utility represents preferences, then higher utility means preferred. That is the *only* sense in which representative utility measures what the agent wants.
- ▶ Nothing requires utility to match *how much* the agent wants an outcome.
- ▶ What utility tracks is the agent’s attitude to **risk**: a much higher utility means she will accept even small probabilities of that outcome. That is all it means.

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- ▶ Rationality might be more **substantive**. Schwarz suggests one way: if rational egoism were true, it would be irrational ever to do anything that was worse for us.
- ▶ But that is not the only way. We might be required to care about certain things: love, friendship, pleasure, carrot cake. We might be required always to do the morally right thing. Rationality might require going off into the woods to commune with god.
- ▶ We assume neither. Our assumptions about rationality are deliberately minimal, and neutral between these visions.

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- ▶ And u is unique up to unit and zero.

A representation theorem. The numbers are read off the preferences.

Plan

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- ▶ $A \sim B$ — **indifference**: $A \succsim B$ and $B \succsim A$.
- ▶ $\succsim$ is the basic notion. The other two are defined from it.

Preference and Choice

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- ▶ If $B \succ A$: the agent will choose B , and will not choose A .
- ▶ If $A \sim B$: either. Choice can't tell them apart.
- ▶ We won't commit to what preference *is*. Two things matter: it is linked to choice as above, and it is open to rational evaluation.
- ▶ It is defined over every pair of gambles, not just the choices the agent actually faces.

What Gets Ranked

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- ▶ $\succsim$ ranges over **all** gambles — every act in every decision table, including ones nobody has offered.

The Ordering Principles

Reflexivity

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- ▶ Both are among the assumptions of expected utility theory.

Some Facts About the Preference Order

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- ▶ Can $A \succ B$ and $B \succ A$ both hold? No. $A \succ B$ says not
 $B \succsim A$.

Some Facts About the Preference Order

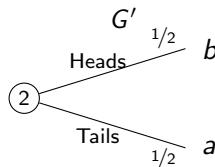
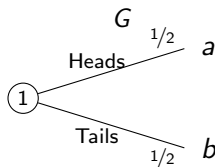
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 $A \succ C$.
- ▶ Can $A \succ B$ and $B \succ A$ both hold? No. $A \succ B$ says not
 $B \succsim A$.
- ▶ Each follows from the definitions plus transitivity. We'll use facts like these without comment.

Two Gambles, Same Chances

	Heads ($1/2$)	Tails ($1/2$)
G	a	b
G'	b	a

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If two gambles give every final outcome the same probability, then the agent is indifferent between them.

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If two gambles give every final outcome the same probability, then the agent is indifferent between them.

- ▶ Only the **chances of the final outcomes** matter. Not which states carry them.

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- ▶ Caspar Hare, “Take the Sugar”, *Analysis* 70 (2010).
- ▶ Tomi Francis and Johan Gustafsson, “Stochastic Dominance for Incomplete Preferences”, *Analysis* 86 (2026), 299–311.

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Definition

The **prospect of a gamble** G , written P_G , is the function taking each outcome x to the probability that G results in x — computed from your credences in the states.

Same Prospect, Different Gambles

	Heads ($1/2$)	Tails ($1/2$)
G	a	b
G'	b	a

$$P_G(a) = 1/2, \quad P_G(b) = 1/2$$

$$P_{G'}(a) = 1/2, \quad P_{G'}(b) = 1/2$$

Same Prospect, Different Gambles

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$$P_G(a) = 1/2, \quad P_G(b) = 1/2$$

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► $P_G = P_{G'}$: the same prospect.

Same Prospect, Different Gambles

	Heads ($1/2$)	Tails ($1/2$)
G	a	b
G'	b	a

$$P_G(a) = 1/2, \quad P_G(b) = 1/2$$

$$P_{G'}(a) = 1/2, \quad P_{G'}(b) = 1/2$$

- ▶ $P_G = P_{G'}$: the same prospect.
- ▶ Stochastic equivalence says exactly that $G \sim G'$.

From Gambles to Prospects

Proposition

Suppose $\succsim$ satisfies stochastic equivalence and transitivity. If $P_G = P_{G'}$ and $P_H = P_{H'}$, then $G \succsim H$ iff $G' \succsim H'$.

$$G' \sim G, \quad H \sim H'$$

Stochastic equivalence: same prospect, so indifferent.

From Gambles to Prospects

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Suppose $G \succsim H$.

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$$G' \succsim G \succsim H$$

$$G' \sim G \text{ gives } G' \succsim G.$$

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Transitivity. And $H \sim H'$ gives $H \succsim H'$.

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Transitivity again. The other direction is the same argument with the primes swapped.

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$$G' \succsim H'$$

Which gambles carry the prospects makes no difference.

Preferences Over Prospects

Definition

For prospects P and Q : $P \succsim Q$ iff $G \succsim H$ for some gambles G, H with $P_G = P$ and $P_H = Q$.

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- ▶ Assumed, not proved here: if the ordering properties hold of $\succsim$ over gambles then, given stochastic equivalence, they hold of $\succsim$ over prospects.

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- ▶ Assumed, not proved here: if the ordering properties hold of $\succsim$ over gambles then, given stochastic equivalence, they hold of $\succsim$ over prospects.
- ▶ Also in the background: every prospect is the prospect of *some* gamble.

Mixing Prospects

Definition

For prospects A and B and $p \in [0, 1]$, $A \# B$ is the prospect with

$$(A \# B)(x) = p \cdot A(x) + (1 - p) \cdot B(x) \quad \text{for each outcome } x.$$

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Mixing Prospects

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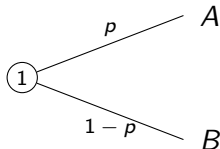
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- ▶ “ A with probability p , otherwise B .”
- ▶ A and B need not be single outcomes. They can themselves be prospects.

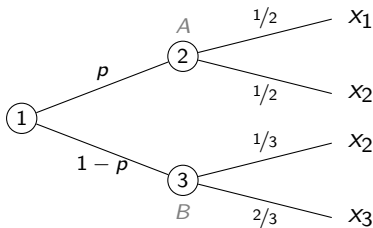
Mixing, in a Picture

$A \succ p B$, where A and B are prospects.



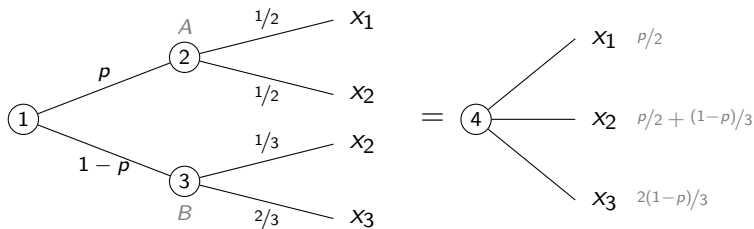
Mixing, in a Picture

Open up A and B : each is itself a spread of chances over outcomes.



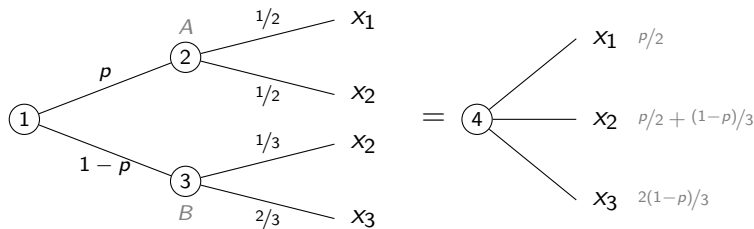
Mixing, in a Picture

Multiply through, and it is one prospect over the outcomes.



Mixing, in a Picture

Multiply through, and it is one prospect over the outcomes.



- ▶ This is an **equation**, not a preference. Prospects with the same chances *are* the same prospect.

The Mixture Space

- ▶ The set of all prospects over the outcomes, together with the mixing operation, is a **mixture space**.

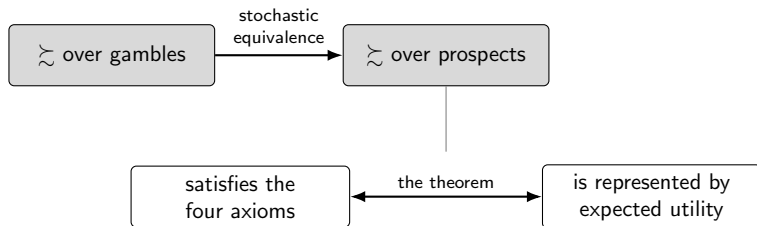
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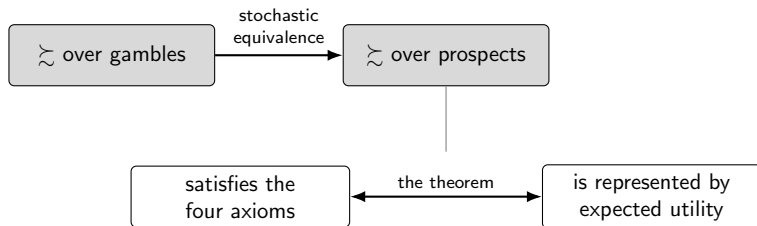
The Mixture Space

- ▶ The set of all prospects over the outcomes, together with the mixing operation, is a **mixture space**.
- ▶ From here on, $\succsim$ is a relation over prospects. This is justified by the Stochastic Equivalence assumption.
- ▶ Single outcomes can be identified with prospects too: x with the prospect that gives x for certain.

What We Are Going to Prove

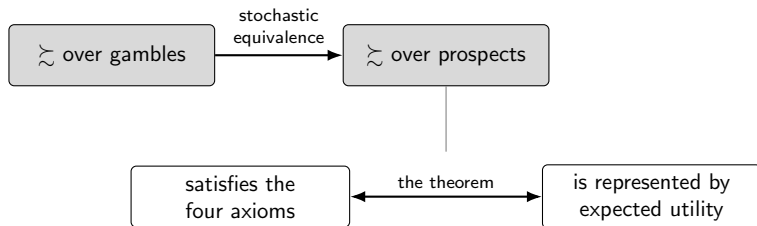


What We Are Going to Prove



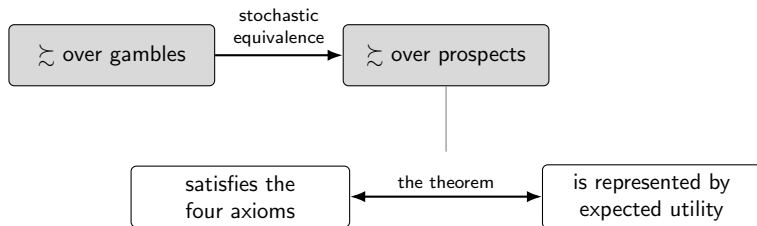
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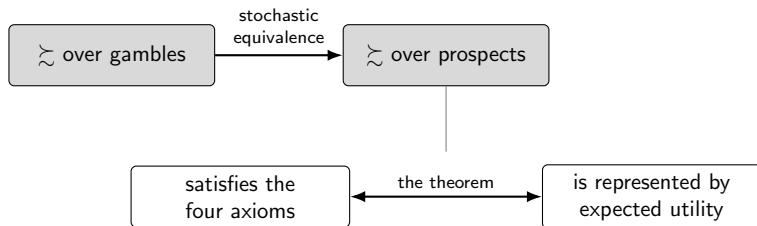
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- ▶ *Represented*: $P \succsim Q$ iff $EU(P) \geq EU(Q)$, for every pair of prospects. The ranking by $\succsim$ and the ranking by expected utility are the same ranking.

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- ▶ *Represented*: $P \succsim Q$ iff $EU(P) \geq EU(Q)$, for every pair of prospects. The ranking by $\succsim$ and the ranking by expected utility are the same ranking.
- ▶ Two axioms so far: reflexivity and transitivity. Three to come.

Completeness

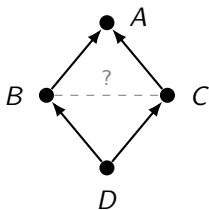
For all prospects A and B : $A \succsim B$ or $B \succsim A$.

Completeness

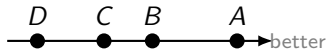
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Without it



With it

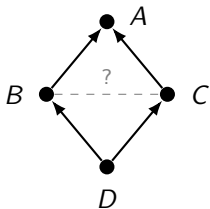


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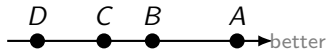
Completeness

For all prospects A and B : $A \succsim B$ or $B \succsim A$.

Without it



With it



- ▶ No gaps: every pair is comparable. With transitivity, everything lines up.

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For all prospects A, B, C and every $p \in (0, 1)$: $A \succsim B$ iff $A p C \succsim B p C$ iff $C p A \succsim C p B$.

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	$E \quad (\text{Cr}(E) = p)$	$\neg E \quad (1 - p)$
$A p C$	A	C
$B p C$	B	C

- ▶ One way to realise them: gambles that agree if $\neg E$. But any gambles with these prospects will do; they need not agree anywhere.

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- ▶ One way to realise them: gambles that agree if $\neg E$. But any gambles with these prospects will do; they need not agree anywhere.
- ▶ Apply it twice: $A p C \succsim B p C$ iff $A \succsim B$ iff $A q C \succsim B q C$.
So the weight makes no difference.

Independence and the Sure-Thing Principle

Sure-thing principle

Gambles G, G', H, H' and an event E . If G, H agree if E and so do G', H' , while G, G' agree if $\neg E$ and so do H, H' : then $G \succsim G'$ iff $H \succsim H'$.

Two gambles that agree on what happens if $\neg E$.

	E (p)	$\neg E$ ($1 - p$)
G	A	C
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Independence and the Sure-Thing Principle

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The shared outcome, in bold.

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Replace it by another shared outcome, D .

	E (p)	$\neg E$ ($1 - p$)
H	A	D
H'	B	D

Gambles G, G', H, H' and an event E . If G, H agree if E and so do G', H' , while G, G' agree if $\neg E$ and so do H, H' : then $G \precsim G'$ iff $H \precsim H'$.

The sure-thing principle: the two comparisons come out the same way.

	$E \quad (p)$	$\neg E \quad (1-p)$
G	A	C
G'	B	C

$G \succsim G' \quad \text{iff} \quad H \succsim H'$

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$$G \sim G' \quad \text{iff} \quad H \sim H'$$

- ▶ A principle about *states*: what the two gambles share cannot affect the comparison.

Independence and the Sure-Thing Principle

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- ▶ A principle about *states*: what the two gambles share cannot affect the comparison.
- ▶ Independence implies it. With stochastic equivalence the two come to the same thing, at least when there are finitely many states.

Continuity

Continuity

If $A \succ B \succ C$, then there is some $p \in (0, 1)$ with $B \sim A p C$.

B sits strictly between A and C .

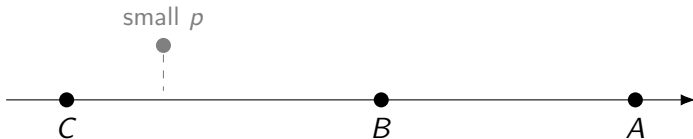


Continuity

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If $A \succ B \succ C$, then there is some $p \in (0, 1)$ with $B \sim ApC$.

$A p C$ runs from C (at $p = 0$) to A (at $p = 1$).

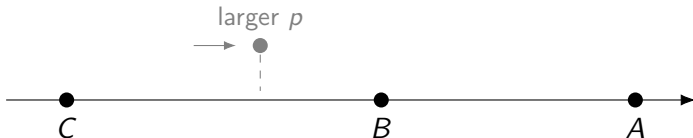


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If $A \succ B \succ C$, then there is some $p \in (0, 1)$ with $B \sim A p C$.

Somewhere on the way it must draw level with B .

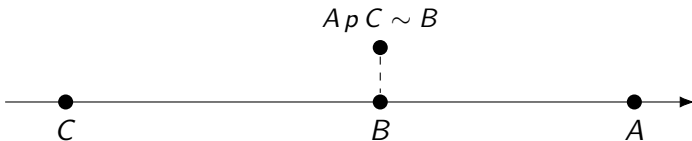


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If $A \succ B \succ C$, then there is some $p \in (0, 1)$ with $B \sim ApC$.

That p is what Continuity promises.

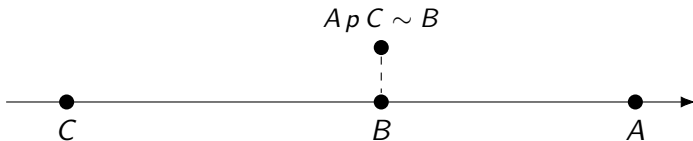


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- No prospect is *infinitely* better than another.

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- ▶ An agent ranks $b \succ c \succ a$, but prefers c to every mixture of b and a , however heavily weighted towards b .

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- ▶ Then there is no p to find, and no number to assign to c .
- ▶ The question is whether such preferences could be **rational**.

The Axioms

Reflexivity

$$A \succsim A$$

Transitivity

$$A \succsim B \text{ and } B \succsim C \text{ give } A \succsim C$$

Completeness

$$A \succsim B \text{ or } B \succsim A$$

Independence

$$A \succsim B \text{ iff } A p C \succsim B p C \text{ iff } C p A \succsim C p B$$

Continuity

$$A \succ B \succ C \text{ gives some } p \text{ with } B \sim A p C$$

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- Reflexivity follows from Completeness: take B to be A . So we can drop it. **Four axioms.**

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- ▶ Reflexivity follows from Completeness: take B to be A . So we can drop it. **Four axioms.**
- ▶ Stochastic equivalence is assumed in the background.

The Expected Utility Theorem

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Let $\succsim$ be a preference relation over prospects.

- (i) $\succsim$ satisfies the four axioms **if and only if** some function u on outcomes represents it by expected utility: for all prospects P and Q ,

$$P \succsim Q \quad \text{iff} \quad \text{EU}(P) \geq \text{EU}(Q),$$

where $\text{EU}(P) = \sum_x P(x) u(x)$, the sum running over the finitely many outcomes P gives positive probability to.

- (ii) Any two such functions u and u' differ only in unit and zero: $u' = a \cdot u + c$ for some $a > 0$ and some c .

The Idea of the Proof

- Pick two prospects, one better than the other: $A_1 \succ A_0$. The choice is more or less arbitrary.

The Idea of the Proof, Continued

- ▶ If $B \succ A_1$: there is exactly one p with $A_1 \sim B p A_0$. Assign $u(B) = 1/p$.

The Idea of the Proof, Continued

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- ▶ The other direction: preferences that violate any one of the axioms cannot be represented by a utility function.

Four Facts About Mixing

$$(M1) \quad A p A = A.$$

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- ▶ All four are arithmetic. Compute the chance of each outcome on both sides.
- ▶ The proof leans on (M1) and (M4).

Beginning the Proof

- ▶ The proof follows Fishburn, *The Foundations of Expected Utility* (1982), ch. 2, Theorem 1. The same proof is Theorem 8.4 of his *Utility Theory for Decision Making* (1970).

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- ▶ If no prospect is strictly preferred to any other, set u constant. Done.
- ▶ Otherwise fix two prospects $A_1 \succ A_0$: the **unit** and the **zero**.

Step 1: More of the Better Thing Is Better

Lemma 1

If $A \succ B$ and $p, q \in [0, 1]$, then $A p B \succ A q B$ iff $p > q$.



Step 1: More of the Better Thing Is Better

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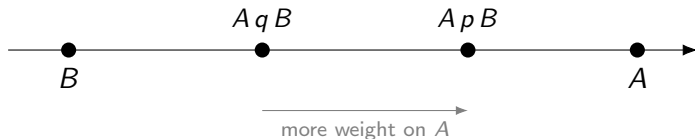
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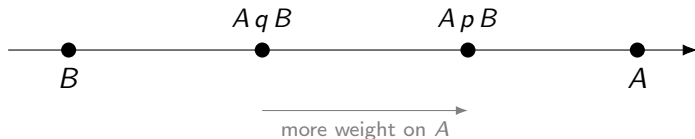
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If $A \succ B$ and $p, q \in [0, 1]$, then $A p B \succ A q B$ iff $p > q$.



- ▶ Sliding weight from B onto A moves you up.

Proving Lemma 1

Given $A \succ B$ and $0 < q < p < 1$ (the edge cases $q = 0$ and $p = 1$ take one step each). Write $s = (p-q)/(1-q)$, so that $s + (1-s)q = p$.

$$A \succ B$$

The hypothesis.

Proving Lemma 1

Given $A \succ B$ and $0 < q < p < 1$ (the edge cases $q = 0$ and $p = 1$ take one step each). Write $s = (p-q)/(1-q)$, so that $s + (1-s)q = p$.

$$A \text{ } q \text{ } A \succ A \text{ } q \text{ } B$$

Independence, mixing A in front of both sides at weight q .

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$$A \succ A q B$$

(M1) on the left.

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$$A \succ (A \ q \ B) \succ (A \ q \ B) \succ (A \ q \ B)$$

Independence again, mixing both sides with $A \ q \ B$ at weight s .

Proving Lemma 1

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$$As(AqB) \succ AqB$$

(M1) on the right.

Proving Lemma 1

Given $A \succ B$ and $0 < q < p < 1$ (the edge cases $q = 0$ and $p = 1$ take one step each). Write $s = (p-q)/(1-q)$, so that $s + (1-s)q = p$.

$$A [s + (1-s)q] B \succ A q B$$

(M4) on the left, with s in the role of p .

Proving Lemma 1

Given $A \succ B$ and $0 < q < p < 1$ (the edge cases $q = 0$ and $p = 1$ take one step each). Write $s = (p-q)/(1-q)$, so that $s + (1-s)q = p$.

$$A p B \succ A q B$$

$s + (1-s)q = p$, by the choice of s .

Proving Lemma 1

Given $A \succ B$ and $0 < q < p < 1$ (the edge cases $q = 0$ and $p = 1$ take one step each). Write $s = (p-q)/(1-q)$, so that $s + (1-s)q = p$.

$$A p B \succ A q B$$

One direction of Lemma 1. For the other: if $p = q$ the two sides are the same prospect, and if $p < q$ the argument just given puts $A q B$ strictly above $A p B$. Either way, not $A p B \succ A q B$.

Step 2: A Mixture of Worse (or Equal) Prospects Is Worse (or Equal)

Lemma 2

If $A \succsim X$ and $A \succsim Y$, then $A \succsim X p Y$.

$$A \succsim X$$

The first hypothesis.

Step 2: A Mixture of Worse (or Equal) Prospects Is Worse (or Equal)

Lemma 2

If $A \succsim X$ and $A \succsim Y$, then $A \succsim X p Y$.

$$A p Y \succsim X p Y$$

Independence, mixing both sides with Y at weight p .

Step 2: A Mixture of Worse (or Equal) Prospects Is Worse (or Equal)

Lemma 2

If $A \sim X$ and $A \sim Y$, then $A \sim X \vee Y$.

$$A p A \rightsquigarrow A p Y$$

Independence, mixing A in front of both sides at weight p .

Step 2: A Mixture of Worse (or Equal) Prospects Is Worse (or Equal)

Lemma 2

If $A \sim X$ and $A \sim Y$, then $A \sim X \dot{\cup} Y$.

$$\mathbf{A} \preceq A_p Y$$

(M1) on the left.

Step 2: A Mixture of Worse (or Equal) Prospects Is Worse (or Equal)

Lemma 2

If $A \succsim X$ and $A \succsim Y$, then $A \succsim X p Y$.

$$A \succsim A p Y \quad \text{and} \quad A p Y \succsim X p Y$$

Put the two results together.

Step 2: A Mixture of Worse (or Equal) Prospects Is Worse (or Equal)

Lemma 2

If $A \succsim X$ and $A \succsim Y$, then $A \succsim X p Y$.

$$A \succsim X p Y$$

Transitivity.

Step 2: A Mixture of Worse (or Equal) Prospects Is Worse (or Equal)

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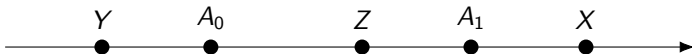
$$A \succsim X p Y$$

A mixture of prospects that A is at least as good as is itself one that A is at least as good as. The same proof with the preferences reversed gives the mirror image.

Intervals

Definition

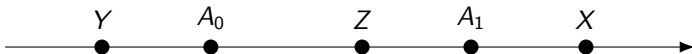
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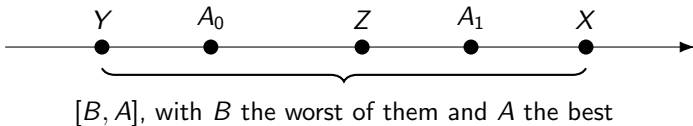


- ▶ Lemma 2: an interval is **closed under mixing**. If X and Y are in $[B, A]$, so is $X \circ Y$.

Intervals

Definition

For prospects $A \succsim B$, the **interval** $[B, A]$ is the set of prospects X with $A \succsim X \succsim B$.



- ▶ Lemma 2: an interval is **closed under mixing**. If X and Y are in $[B, A]$, so is $X \text{ } q \text{ } Y$.
- ▶ Any **finitely many** prospects have a best and a worst among them, by Completeness and Transitivity. So they all lie in one interval.

Step 3: Calibration on an Interval

Lemma 3

Fix $A \succ B$. For every X in $[B, A]$ there is **exactly one** $p \in [0, 1]$ with $X \sim ApB$.

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Step 3: Calibration on an Interval

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- ▶ Otherwise $A \succ X \succ B$, and Continuity hands us a p .

Step 3: Calibration on an Interval

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- **Existence.** If $X \sim A$, take $p = 1$. If $X \sim B$, take $p = 0$.
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- **Uniqueness.** Two different values $p > q$ would give $A p B \succ A q B$, by Lemma 1.

Step 3: Calibration on an Interval

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- ▶ **Uniqueness.** Two different values $p > q$ would give $A p B \succ A q B$, by Lemma 1.

Lemma 1, again

If $A \succ B$ and $p, q \in [0, 1]$, then $A p B \succ A q B$ iff $p > q$.

Step 3: Calibration on an Interval

Lemma 3

Fix $A \succ B$. For every X in $[B, A]$ there is **exactly one** $p \in [0, 1]$ with $X \sim A p B$.

- ▶ **Existence.** If $X \sim A$, take $p = 1$. If $X \sim B$, take $p = 0$.
- ▶ Otherwise $A \succ X \succ B$, and Continuity hands us a p .
- ▶ **Uniqueness.** Two different values $p > q$ would give $A p B \succ A q B$, by Lemma 1.
- ▶ But both are $\sim X$. Contradiction, by transitivity.

Defining u_{AB}

Definition

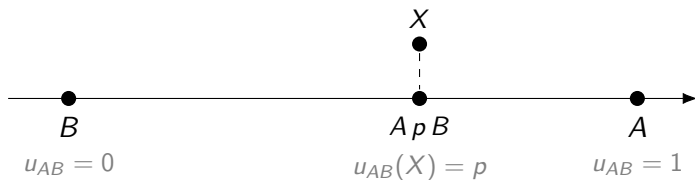
For X in $[B, A]$: $u_{AB}(X)$ is the unique p with $X \sim A p B$.



Defining u_{AB}

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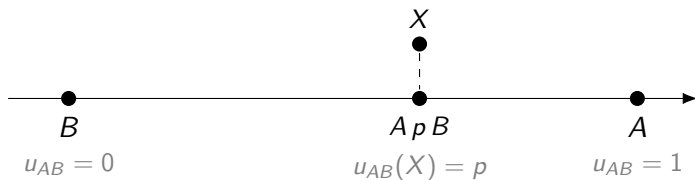
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Defining u_{AB}

Definition

For X in $[B, A]$: $u_{AB}(X)$ is the unique p with $X \sim A p B$.



- One calibration per interval. The subscript keeps track of which ends were used.

Step 4: u_{AB} Represents $\succsim$ on the Interval

Lemma 4

For X and Y in $[B, A]$: $X \succsim Y$ iff $u_{AB}(X) \geq u_{AB}(Y)$.

Write $p = u_{AB}(X)$ and $q = u_{AB}(Y)$, so that $X \sim ApB$ and $Y \sim AqB$.

$$X \succsim Y$$

Start from the left-hand side.

Step 4: u_{AB} Represents $\succsim$ on the Interval

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For X and Y in $[B, A]$: $X \succsim Y$ iff $u_{AB}(X) \geq u_{AB}(Y)$.

Write $p = u_{AB}(X)$ and $q = u_{AB}(Y)$, so that $X \sim ApB$ and $Y \sim AqB$.

$$X \succsim Y \quad \text{iff} \quad \mathbf{ApB} \succsim \mathbf{AqB}$$

Swap X and Y for the prospects they are indifferent to. Transitivity, in both directions.

Step 4: u_{AB} Represents $\succsim$ on the Interval

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For X and Y in $[B, A]$: $X \succsim Y$ iff $u_{AB}(X) \geq u_{AB}(Y)$.

Write $p = u_{AB}(X)$ and $q = u_{AB}(Y)$, so that $X \sim A p B$ and $Y \sim A q B$.

$$X \succsim Y \quad \text{iff} \quad \text{not } A q B \succ A p B$$

Completeness: $P \succsim Q$ holds exactly when $Q \succ P$ fails.

Step 4: u_{AB} Represents $\succsim$ on the Interval

Lemma 4

For X and Y in $[B, A]$: $X \succsim Y$ iff $u_{AB}(X) \geq u_{AB}(Y)$.

Write $p = u_{AB}(X)$ and $q = u_{AB}(Y)$, so that $X \sim A p B$ and $Y \sim A q B$.

$$X \succsim Y \quad \text{iff} \quad \text{not } \mathbf{q} > \mathbf{p}$$

Lemma 1, in both directions: $A q B \succ A p B$ iff $q > p$.

Step 4: u_{AB} Represents $\succsim$ on the Interval

Lemma 4

For X and Y in $[B, A]$: $X \succsim Y$ iff $u_{AB}(X) \geq u_{AB}(Y)$.

Write $p = u_{AB}(X)$ and $q = u_{AB}(Y)$, so that $X \sim ApB$ and $Y \sim AqB$.

$$X \succsim Y \quad \text{iff} \quad p \geq q$$

And $p \geq q$ is $u_{AB}(X) \geq u_{AB}(Y)$.

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For X and Y in $[B, A]$: $X \succsim Y$ iff $u_{AB}(X) \geq u_{AB}(Y)$.

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$$X \succsim Y \quad \text{iff} \quad p \geq q$$

Which is Lemma 4. Every step was an iff.

Step 5: u_{AB} Respects Mixing on the Interval

Lemma 5

For X and Y in $[B, A]$ and $p \in [0, 1]$:

$$u_{AB}(X \, p \, Y) = p \cdot u_{AB}(X) + (1 - p) \cdot u_{AB}(Y).$$

Write $q = u_{AB}(X)$ and $r = u_{AB}(Y)$, so that $X \sim AqB$ and $Y \sim ArB$.

$$X \, p \, Y$$

Start with the mixture. It is in $[B, A]$ by Lemma 2, so u_{AB} is defined on it.

Step 5: u_{AB} Respects Mixing on the Interval

Lemma 5

For X and Y in $[B, A]$ and $p \in [0, 1]$:

$$u_{AB}(X \ p \ Y) = p \cdot u_{AB}(X) + (1 - p) \cdot u_{AB}(Y).$$

Write $q = u_{AB}(X)$ and $r = u_{AB}(Y)$, so that $X \sim A \ q \ B$ and $Y \sim A \ r \ B$.

$$X \ p \ Y \sim (A \ q \ B) \ p \ Y$$

$X \sim A \ q \ B$. Independence, mixing Y behind both at weight p .

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$$X \, p \, Y \sim (A \, q \, B) \, p \, (A \, r \, B)$$

$Y \sim ArB$. Independence, mixing AqB in front of both at weight p .

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Write $q = u_{AB}(X)$ and $r = u_{AB}(Y)$, so that $X \sim AqB$ and $Y \sim ArB$.

$$X \, p \, Y \sim A[pq + (1 - p)r]B$$

Arithmetic: a mixture of two A – B mixtures is an A – B mixture, with the weights averaged.

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$$u_{AB}(X \, p \, Y) = pq + (1 - p)r$$

Lemma 3: that is the unique calibrating value.

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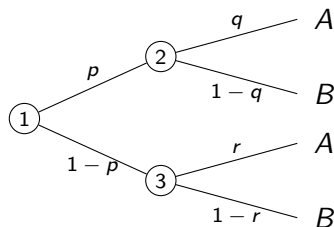
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Which is Lemma 5.

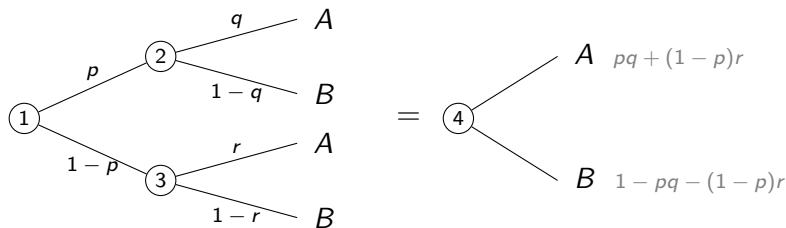
Lemma 5, in a Picture

$X \succsim_p Y$, with X and Y replaced by their calibrations.



Lemma 5, in a Picture

Multiply through: A with total weight $pq + (1 - p)r$, B with the rest.



Rescaling Keeps Both Properties

Suppose g represents $\succsim$ and respects mixing on some interval. Take $a > 0$ and any c , and put $h = a \cdot g + c$.

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- ▶ h **respects mixing**. Using $c = pc + (1 - p)c$:

$$\begin{aligned}
 h(X \, p \, Y) &= a[p g(X) + (1 - p) g(Y)] + c \\
 &= p[ag(X) + c] + (1 - p)[ag(Y) + c] \\
 &= p h(X) + (1 - p) h(Y).
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- ▶ h **respects mixing**. Using $c = pc + (1 - p)c$:

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 h(X \text{ } p \text{ } Y) &= a[p g(X) + (1 - p) g(Y)] + c \\
 &= p[ag(X) + c] + (1 - p)[ag(Y) + c] \\
 &= p h(X) + (1 - p) h(Y).
 \end{aligned}$$

- ▶ So a rescaling with a positive factor loses nothing. We use this twice: next, and for uniqueness.

The Five Lemmas

Lemma 1

If $A \succ B$ and $p, q \in [0, 1]$, then $A p B \succ A q B$ iff $p > q$.

Lemma 2

If $A \succsim X$ and $A \succsim Y$, then $A \succsim X p Y$; likewise with $\succsim$ reversed. So an interval is closed under mixing.

Lemma 3

Fix $A \succ B$. For every X in $[B, A]$ there is exactly one $p \in [0, 1]$ with $X \sim A p B$. That p is $u_{AB}(X)$.

Lemma 4

For X and Y in $[B, A]$: $X \succsim Y$ iff $u_{AB}(X) \geq u_{AB}(Y)$.

Lemma 5

For X and Y in $[B, A]$: $u_{AB}(X p Y) = p \cdot u_{AB}(X) + (1 - p) \cdot u_{AB}(Y)$.

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Definition

$$f(X) = \frac{u_{AB}(X) - u_{AB}(A_0)}{u_{AB}(A_1) - u_{AB}(A_0)} \quad \text{for } X \text{ in } [B, A].$$

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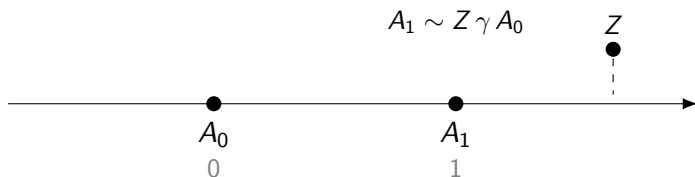
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- ▶ f is a rescaling of u_{AB} with a positive factor, so it still represents $\succsim$ and respects mixing on $[B, A]$: Lemmas 4 and 5, plus the previous slide.
- ▶ Now three cases show what $f(Z)$ must be — **without reference to the interval**.

The Three Cases

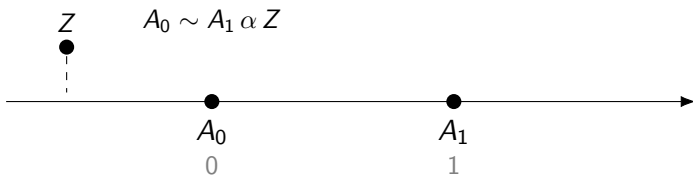
Case 2: $Z \succ A_1$. Lemma 3 on $[A_0, Z]$: $A_1 \sim Z \gamma A_0$, for exactly one γ .



- ▶ Case 1: $f(Z) = \beta \cdot 1 + (1 - \beta) \cdot 0 = \beta$.
- ▶ Case 2: $1 = f(A_1) = \gamma f(Z) + (1 - \gamma) \cdot 0$, so $f(Z) = 1/\gamma$. And $\gamma > 0$, since $A_1 \succ A_0$.

The Three Cases

Case 3: $A_0 \succ Z$. Lemma 3 on $[Z, A_1]$: $A_0 \sim A_1 \alpha Z$, for exactly one α .



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- ▶ Case 3: $0 = f(A_0) = \alpha \cdot 1 + (1 - \alpha) f(Z)$, so $f(Z) = -\alpha/(1-\alpha)$. And $\alpha < 1$, likewise.

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- ▶ Define $u(Z)$ to be that common value.

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This is exactly the assignment from the Idea of the Proof: β
in between, $1/\gamma$ above, $-\alpha/(1-\alpha)$ below.

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- ▶ On that interval u is the rescaled calibration f , which represents $\succsim$ and respects mixing there.
- ▶ So $X \succsim Y$ iff $u(X) \geq u(Y)$, and $u(X \ q \ Y) = q u(X) + (1 - q) u(Y)$.

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- ▶ A prospect X gives positive probability to finitely many outcomes, say $x_1, \dots, x_n$, with probabilities $p_1, \dots, p_n$.

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- ▶ u respects mixing, so applying this $n - 1$ times:

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$$u(X) = p_1 u(x_1) + p_2 u(x_2) + \dots + p_n u(x_n) = \text{EU}(X).$$

- ▶ Since u represents $\succsim$: $X \succsim Y$ iff $\text{EU}(X) \geq \text{EU}(Y)$. **The representation.**

Step 8: Uniqueness

Suppose v also represents $\succsim$ and respects mixing. We want $a > 0$ and c with $v = a \cdot u + c$.

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- ▶ **What is left:** show $v(Z) = w(Z)$ for every Z . Suppose not, for some Z . Three cases, each a contradiction.

Uniqueness: The Three Cases

v and w both represent $\succsim$ (so indifferent prospects get equal values),
both respect mixing, and they agree at A_1 and A_0 .

- **Case 1:** $A_1 \succsim Z \succsim A_0$, so $Z \sim A_1 \beta A_0$. Then $v(Z) = \beta v(A_1) + (1 - \beta) v(A_0) = \beta w(A_1) + (1 - \beta) w(A_0) = w(Z)$.

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- ▶ **Case 2:** $Z \succ A_1$, so $A_1 \sim Z \gamma A_0$ with $\gamma > 0$. Then $\gamma v(Z) + (1 - \gamma) v(A_0) = v(A_1) = w(A_1) = \gamma w(Z) + (1 - \gamma) w(A_0)$, so $v(Z) = w(Z)$.

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- ▶ **Case 3:** $A_0 \succ Z$, so $A_0 \sim A_1 \alpha Z$ with $\alpha < 1$. Then $\alpha v(A_1) + (1 - \alpha) v(Z) = v(A_0) = w(A_0) = \alpha w(A_1) + (1 - \alpha) w(Z)$, so $v(Z) = w(Z)$.

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- ▶ Contradiction in every case. **So** $v = w = a \cdot u + c$, **with** $a = v(A_1) - v(A_0) > 0$ **and** $c = v(A_0)$.

Rescaled Expected Utility Is Expected Utility

Take any $a > 0$ and c , and put $u' = a \cdot u + c$.

$$EU'(P) = \sum_x P(x) u'(x)$$

Expected utility computed with u' .

Rescaled Expected Utility Is Expected Utility

Take any $a > 0$ and c , and put $u' = a \cdot u + c$.

$$EU'(P) = \sum_x P(x) [a u(x) + c]$$

Substitute $u' = a u + c$.

Rescaled Expected Utility Is Expected Utility

Take any $a > 0$ and c , and put $u' = a \cdot u + c$.

$$EU'(P) = a \sum_x P(x) u(x) + c \sum_x P(x)$$

Collect the a terms and the c terms.

Rescaled Expected Utility Is Expected Utility

Take any $a > 0$ and c , and put $u' = a \cdot u + c$.

$$EU'(P) = a \sum_x P(x) u(x) + c \cdot \mathbf{1}$$

The probabilities sum to 1.

Rescaled Expected Utility Is Expected Utility

Take any $a > 0$ and c , and put $u' = a \cdot u + c$.

$$EU'(P) = a \cdot EU(P) + c$$

An increasing linear function of EU.

Rescaled Expected Utility Is Expected Utility

Take any $a > 0$ and c , and put $u' = a \cdot u + c$.

$$EU'(P) = a \cdot EU(P) + c$$

Since $a > 0$: $EU'(P) \geq EU'(Q)$ exactly when $EU(P) \geq EU(Q)$.

Step 9: The Other Direction

Suppose $\succsim$ is represented by expected utility for some u . Then each axiom holds.

► Transitivity and Completeness.

$\geq$ on numbers is transitive and complete, and $\succsim$ copies it: $P \succsim Q$ iff $EU(P) \geq EU(Q)$.

Step 9: The Other Direction

Suppose $\succsim$ is represented by expected utility for some u . Then each axiom holds.

- ▶ **Transitivity and Completeness.**
- ▶ **Independence.**

$EU(A \ p \ C) = p EU(A) + (1 - p) EU(C)$. So

$EU(A \ p \ C) \geq EU(B \ p \ C)$ iff $p EU(A) \geq p EU(B)$ iff $EU(A) \geq EU(B)$, since $p > 0$.

The same for $C \ p \ A$ and $C \ p \ B$, since $1 - p > 0$.

Step 9: The Other Direction

Suppose $\succsim$ is represented by expected utility for some u . Then each axiom holds.

- ▶ **Transitivity and Completeness.**
- ▶ **Independence.**
- ▶ **Continuity.**

Given $EU(A) > EU(B) > EU(C)$, put $p = [EU(B) - EU(C)] / [EU(A) - EU(C)]$, which lies in $(0, 1)$. Then

$$EU(A \ p \ C) = EU(C) + p[EU(A) - EU(C)] = EU(B),$$

so $B \sim A \ p \ C$.

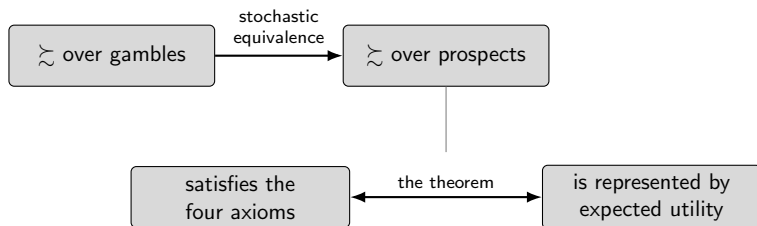
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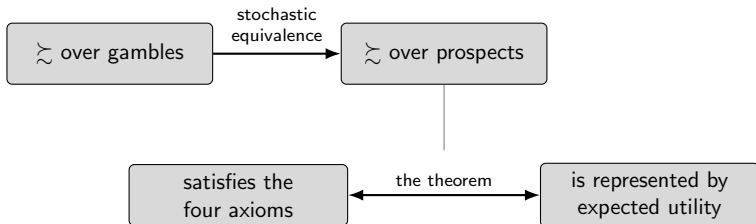
- ▶ **Transitivity and Completeness.**
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Both directions of (i), and (ii). The theorem is proved.

What We Have Shown

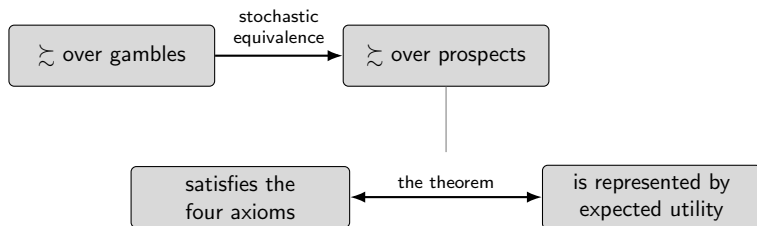


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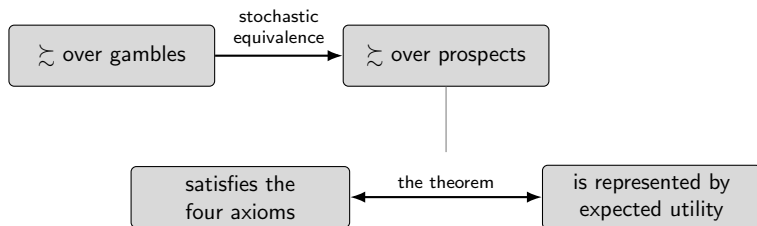
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- ▶ Four axioms on preference hold exactly when some utility function on outcomes represents it — however many outcomes there are.
- ▶ Preference between prospects is exactly comparison of expected utility.
- ▶ The function is fixed once the unit and zero are: an interval scale.

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- ▶ The theorem says the preferences *can be described as* expected-utility maximising.
- ▶ It does not say the agent computes anything. It does not say u is a quantity in her head that *causes* the preferences.

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- ▶ Not: “calculate expectations and choose the biggest”.
- ▶ Rather: have preferences satisfying the **axioms**.
- ▶ Maximising expected utility is then a *consequence*, not an instruction.

How Many Utility Functions?

- ▶ The theorem says **some** u exists. It picks out no particular one.

How Many Utility Functions?

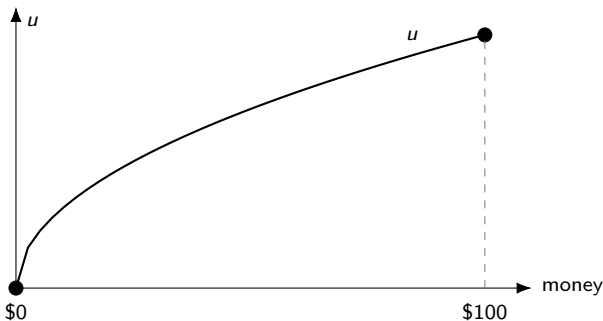
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How Many Utility Functions?

- ▶ The theorem says **some** u exists. It picks out no particular one.
- ▶ Every $a \cdot u + c$ with $a > 0$ does the same job. Nothing else does.
- ▶ Ratios of *differences* do mean something: they survive every rescaling.

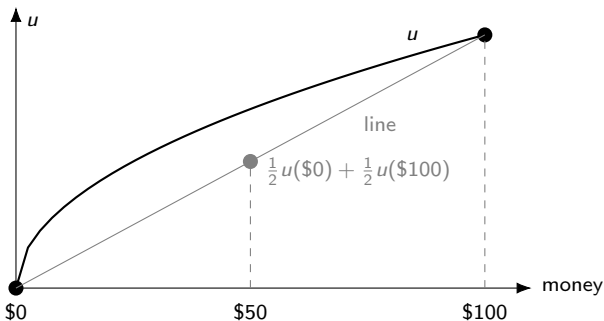
Risk Aversion: Two Senses

The black curve is u , the utility of money. A fair coin for \$100 or \$0, against \$50 for sure.



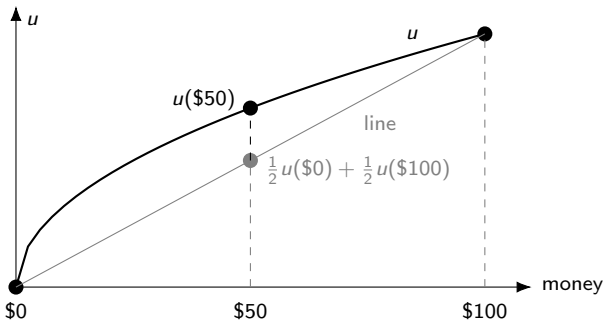
Risk Aversion: Two Senses

The grey line joins $u(\$0)$ to $u(\$100)$. Its midpoint is the gamble's expected utility.



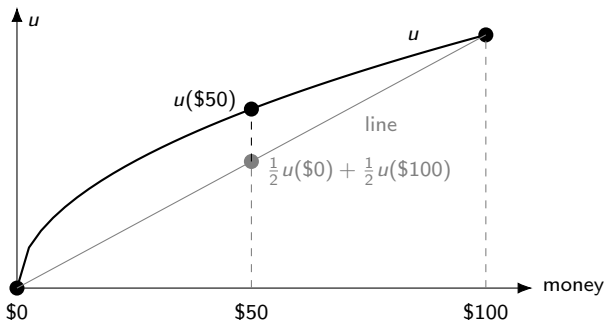
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Risk Aversion: Two Senses

A concave u puts $u(\$50)$ **above** the line: the sure thing wins. Risk aversion about money, inside EUT.



- ▶ Risk aversion about **utility itself** — preferring a sure thing to a gamble with the same *expected utility* — is ruled out.

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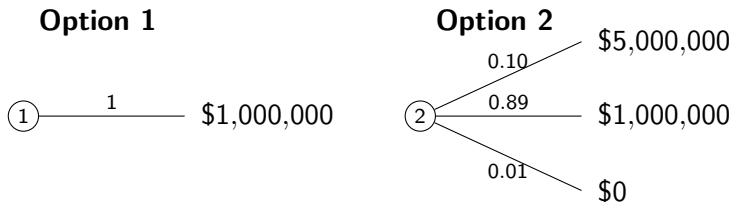
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The Axioms, Again

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- ▶ **Independence.** Now: the Allais paradox.

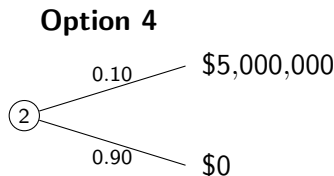
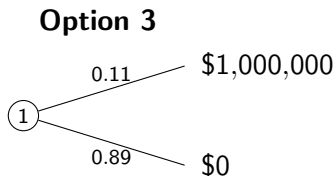
A Choice

Which would you take?



Another Choice

And now?



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- Write Z for: \$5,000,000 with probability $10/11$, else \$0.

	with probability 0.11	with probability 0.89
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Option 2	Z	\$1,000,000
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- ▶ Within each pair, the second column is **shared**.

What Independence Says About It

Write M for \$1,000,000 for certain, and O for \$0 for certain.

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- Independence: Option 1 $\succsim$ Option 2 iff $M \succsim Z$ iff Option 3 $\succsim$ Option 4.

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Option 1 = $M 0.11 M$ Option 3 = $M 0.11 O$

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- ▶ Independence: Option 1 $\succsim$ Option 2 iff $M \succsim Z$ iff Option 3 $\succsim$ Option 4.
- ▶ So Option 1 $\succ$ Option 2 together with Option 4 $\succ$ Option 3 **violates Independence.**

Ellsberg's Urn

30 red balls, and 60 that are black or yellow in an unknown proportion.
One ball is drawn.

	red (30)	black (?)	yellow (?)
Bet I	\$100	\$0	\$0
Bet II	\$0	\$100	\$0
Bet III	\$100	\$0	\$100
Bet IV	\$0	\$100	\$100

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- ▶ **Redescribe the outcomes.** \$0 after turning down a sure million is not the same outcome as \$0. Then Independence isn't violated — but it becomes hard to violate anything.
- ▶ **Give up Independence.** Theories that weight probabilities by rank, or by the agent's attitude to risk.

Next Week

Money pumps and dynamic choice

- ▶ Gustafsson, *Money-Pump Arguments*, chs. 1–2 — on the website
- ▶ BDRC ch. 8